



ARVA TRACKER V2

Product, Monetization & Technical Blueprint

Blueprint lengkap dari monetisasi hingga arsitektur production-ready: Pricing, Database Schema, Coach Mode, Community Mode, AI Coach, Payment Flow, Roadmap MVP 90 Hari, dan Master Prompt Codex.

LANGKAH KECIL, PERUBAHAN BESAR.

01

Monetisasi

Free → Pro → Coach → Community

02

Platform

Multi-tenant, secure, scalable

03

AI Layer

Insight, accountability, retention

Blueprint Arva Tracker V2

Dokumen eksekusi produk, monetisasi, arsitektur, dan roadmap 90 hari

Executive North Star

Arva Tracker V2 tidak diposisikan sebagai habit tracker biasa. Produk ini dirancang sebagai Progress & Accountability Platform untuk individual, coach, komunitas, dan organisasi - dengan monetisasi berlapis dan fondasi SaaS multi-tenant sejak awal.

Outcome bisnis yang ditargetkan

- Membuka recurring revenue dari Personal Pro, Coach Pro, Community, dan Business.
- Meningkatkan retention melalui streak, accountability, weekly report, dan AI insight.
- Menciptakan ARPU lebih tinggi melalui Coach Mode dan Community Mode.
- Menyiapkan jalur marketplace program dan white-label tanpa membebani MVP.
- Membangun data model dan security boundary yang aman untuk multi-tenant.

Keputusan arsitektur utama

Area	Keputusan V2	Alasan
Frontend	Next.js App Router + TypeScript + Tailwind CSS	Iterasi cepat, SSR/RSC, route handlers, DX kuat.
Backend	Next.js server layer + Supabase	MVP cepat; Auth, PostgreSQL, storage, RLS tersedia.
Database	PostgreSQL multi-tenant	Relational, analytics-friendly, mudah di-scale.
Billing	PaymentProvider abstraction	Bisa Midtrans/Xendit tanpa lock-in di domain logic.
AI	AI Gateway + model routing	Kontrol biaya, privacy, dan fallback provider.
Observability	Sentry + audit log + structured logging	Debugging cepat dan trace perubahan sensitif.

Prinsip produk

- User selalu tahu "apa yang harus saya lakukan hari ini?"
- Coach selalu tahu "siapa yang membutuhkan perhatian?"
- Admin selalu tahu "program mana yang aktif dan sehat?"
- Billing dan entitlement selalu diputuskan server-side.
- AI memberi insight, bukan mengambil alih keputusan pengguna.

NAVIGASI DOKUMEN

Daftar Isi & Cara Menggunakan Blueprint

#	Bagian	Fokus
01	Product Vision & Positioning	Positioning dan target persona
02	Monetization Engine	Sumber revenue dan flywheel
03	Paket Harga & Entitlement	Tier, harga, limit fitur
04	Feature Architecture	Modul platform V2
05	Database Schema	Tabel, relasi, index
06	Auth, RLS & Security	Multi-tenant dan hardening
07	Coach Mode	Client management dan intervention
08	Community Mode	Challenge, member, leaderboard
09	AI Coach	Insight, routing, privacy
10	Payment Flow	Checkout, webhook, entitlement
11	Analytics, Retention & Growth	Event tracking dan referral
12	UI/UX Blueprint	Screen, states, UX rules
13	API & Background Jobs	Route handlers, jobs, cron
14	Observability & Operations	Logging, audit, backup
15	Roadmap MVP 90 Hari	12 minggu shipping plan
16	Definition of Done & KPI	Gate release dan metrik
17	Master Prompt Codex	Prompt siap submit di VS Code
18	Appendix Implementasi	ENV, folder tree, checklist

Cara pakai

Bagian 01-16 adalah blueprint keputusan. Bagian 17 adalah master prompt yang bisa langsung ditempel ke Codex. Bagian 18 adalah checklist implementasi untuk mengurangi ambiguity ketika mulai coding.

STRATEGY

01. Product Vision & Positioning

Positioning utama

Arva Tracker

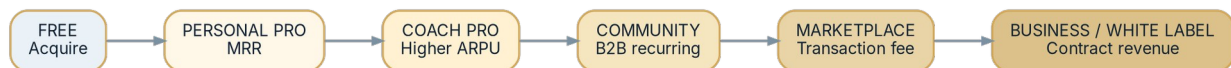
Progress & Accountability Platform - Langkah kecil, perubahan besar.

Nilai jual utama bukan pada kemampuan mencentang aktivitas. Nilai jual terletak pada kemampuan mengubah target menjadi sistem harian yang terukur, menjaga accountability, dan membuat progres dapat dilihat oleh user, coach, atau komunitas sesuai izin.

Persona & job-to-be-done

Persona	Masalah utama	Arva Tracker harus memberi hasil
Individual	Target banyak tetapi progres kabur	Rencana harian jelas, streak, laporan, insight.
Coach/Mentor	Monitoring klien tersebar di chat/sheet	Satu dashboard untuk client progress dan intervention.
Community Admin	Sulit mengukur partisipasi member	Challenge, completion rate, leaderboard, nudges.
Business/Team	Aktivitas tim tidak konsisten	Program terstruktur, role, report, audit, export.

Value ladder



Monetisasi dinaikkan berdasarkan nilai hasil, bukan sekadar jumlah fitur.

North-star product metric

Weekly Successful Progress (WSP): jumlah pengguna aktif yang menyelesaikan minimal 70% aktivitas terjadwal pada minggu berjalan. Metrik ini mengikat activation, retention, dan perceived value dalam satu angka yang lebih bermakna daripada sekadar login.

BUSINESS MODEL

02. Monetization Engine

Revenue streams prioritas

Prioritas	Revenue stream	Kapan dijalankan	Catatan
P1	Personal Pro subscription	MVP	MRR awal, validasi willingness-to-pay.
P1	Coach Pro subscription	MVP+	ARPU lebih tinggi, nilai jelas.
P2	Community subscription	Setelah Coach stabil	B2B2C recurring.
P2	Annual plan	Sejak billing siap	Cashflow lebih sehat, churn turun.
P3	Program marketplace fee	Setelah creator supply ada	10-15% platform fee.
P3	White-label / Business	Setelah multi-tenant matang	Setup + recurring contract.

Monetization flywheel

1. Free user mengalami aha moment melalui progress yang terlihat.
2. Pro membuka analytics, unlimited history, dan AI insights.
3. Coach membawa client baru ke platform - acquisition menjadi B2B2C.
4. Community membawa banyak member sekaligus - CAC per user turun.
5. Marketplace membuat creator membawa audience sendiri dan Arva mengambil transaction fee.

Prinsip paywall

Jangan mengunci core habit loop. User gratis harus bisa merasakan manfaat. Paywall ditempatkan pada scale, analytics, collaboration, AI, export, dan automation.

Target unit economics awal

Metrik	Target 90-180 hari	Catatan
Free → Pro conversion	3-7%	Setelah activation $\geq$ 50%.
Monthly logo churn Personal	< 6%	Dorong annual plan.
Coach monthly churn	< 4%	Nilai harus terikat ke client workflow.
Gross margin	> 75%	Kontrol AI + payment + infra.
LTV:CAC	> 3x	Mulai ukur setelah channel stabil.
Payback period	< 4 bulan	Khusus paid acquisition.

PRICING

03. Paket Harga & Entitlement

Pricing rekomendasi launch

Paket	Bulanan	Tahunan	Target
Free	Rp0	Rp0	Acquisition + activation
Personal Pro	Rp39.000	Rp349.000	Individual serius
Coach Pro	Rp199.000	Rp1.790.000	Coach / mentor
Community	Rp599.000	Rp5.390.000	Komunitas / cohort
Business	Mulai Rp1.500.000	Custom	Team / organisasi

Harga bersifat rekomendasi awal. Gunakan feature flag dan tabel plans agar bisa A/B test tanpa mengubah codebase. Tampilkan harga tahunan sebagai opsi best value, bukan diskon permanen.

Entitlement matrix

Fitur	Free	Pro	Coach	Community
Program aktif	3	Unlimited	Unlimited	Unlimited
History	30 hari	Unlimited	Unlimited	Unlimited
Advanced analytics	-	Ya	Ya	Ya
Weekly AI insight	-	Ya	Ya	Ya
Accountability partner	1	3	Client	Member
Client management	-	-	Hingga 30*	Hingga 100*
Community challenge	-	-	1 cohort	Unlimited*
Export PDF/CSV	-	Ya	Ya	Ya
Custom branding	-	-	Lite	Ya
Priority support	-	-	Ya	Ya

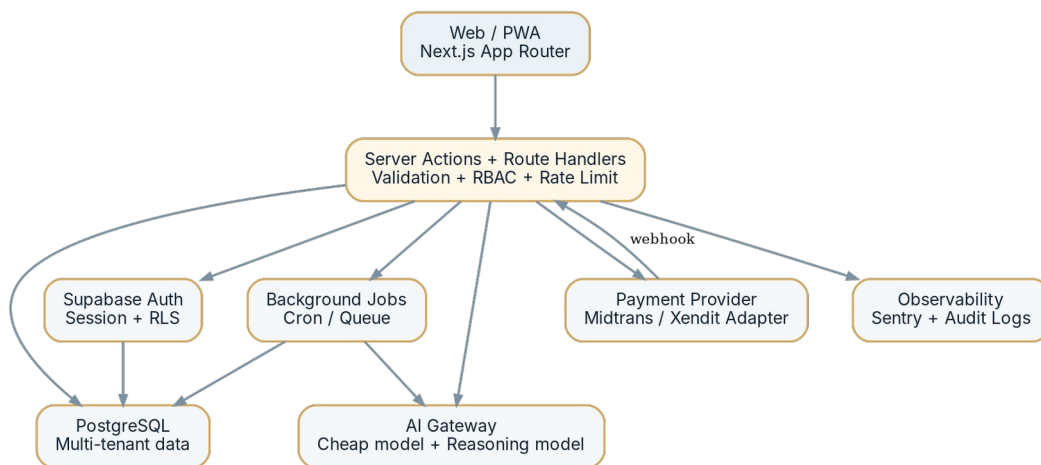
*Limit harus berasal dari database plan entitlements, bukan hard-code. Ini memungkinkan trial, promo, enterprise override, dan paket custom.

Trial & promotional mechanics

- 14 hari Coach Pro trial tanpa kartu untuk acquisition coach.
- Personal Pro trial hanya setelah user mencapai activation event (mis. 7 check-ins).
- Coupon, referral credit, dan trial extension dicatat sebagai ledger - bukan mengubah tanggal subscription secara liar.
- Grace period 3-7 hari ketika renewal gagal, dengan banner non-intrusif dan retry flow.

PRODUCT MODULES

04. Feature Architecture



Arsitektur aplikasi V2: UI tipis, keputusan akses dan billing berada di server.

Modul domain

Domain	Komponen utama	MVP?
Identity	Profile, auth, session, roles	Ya
Workspace	Personal / Coach / Community / Business tenant	Ya
Program	Program, steps, schedule, enrollments	Ya
Tracking	Activity log, check-in, reflection, progress	Ya
Accountability	Partner, coach-client, nudges	Ya
Billing	Plans, subscription, transaction, entitlement	Ya
AI	Insight, prompt version, usage, provider routing	Ya (weekly)
Growth	Referral, share card, invite	P2
Marketplace	Listing, order, payout	P3
White-label	Theme/domain/custom mail	P3

Core domain rule

Satu sumber kebenaran

Semua data kolaboratif harus punya workspace_id. Semua akses tenant wajib melewati membership + RLS. Personal workspace dibuat otomatis untuk setiap user agar model data tetap konsisten.

DATA MODEL

05. Database Schema

Entity model inti

Supabase Auth mengelola auth.users. Aplikasi menggunakan UUID, timestampz UTC, created_at/updated_at, dan soft-delete hanya untuk entitas bisnis yang membutuhkan recovery. Hindari menaruh entitlement sebagai boolean tersebar di profile.

Identity & Workspace

Table	Kolom kunci	Catatan
profiles	id FK auth.users, full_name, avatar_url, timezone, locale	Tidak menyimpan secret.
workspaces	id, type, name, slug, owner_id, status	type: personal/coach/community/business.
workspace_members	workspace_id, user_id, role, status, joined_at	Unique(workspace_id,user_id).

Table	Kolom kunci	Catatan
workspace_invites	id, workspace_id, email/phone_hash, role, token_hash, expires_at	Token mentah tidak disimpan.

Program & Tracking

Table	Kolom kunci	Index/constraint
programs	id, workspace_id, title, description, visibility, status, start_mode	idx workspace,status
program_steps	id, program_id, day_no, order_no, title, kind, config_json	unique program/day/order
program_enrollments	id, program_id, user_id, start_date, status, coach_id	unique active enrollment
activity_logs	id, enrollment_id, step_id, activity_date, status, completed_at, source	idx enrollment,date
daily_reflections	id, enrollment_id, activity_date, note, mood, score	unique enrollment,date
progress_snapshots	id, enrollment_id, period, completion_rate, streak, metrics_json	analytics cache

Coach, Community & Accountability

Table	Kolom kunci	Tujuan
coach_client_links	id, workspace_id, coach_id, client_id, status, started_at	Relasi coach-client eksplisit.
accountability_links	id, owner_id, partner_id, scope, status, permissions_json	Sharing dengan consent.
coach_interventions	id, coach_id, client_id, enrollment_id, type, note, resolved_at	Audit follow-up.
community_challenges	id, workspace_id, program_id, title, start_at, end_at, rules_json	Challenge cohort.
challenge_members	challenge_id, user_id, enrollment_id, joined_at	Unique member/challenge.

Billing & Growth

Table	Kolom kunci	Tujuan
plans	id, code, name, billing_period, price, currency, entitlements_json, active	Catalog server-side.
subscriptions	id, workspace_id, plan_id, provider, provider_ref, status, current_period_end	Sumber akses berbayar.
billing_transactions	id, subscription_id, amount, status, provider_event_id, paid_at	Ledger pembayaran.

Table	Kolom kunci	Tujuan
webhook_events	id, provider, event_id, payload_hash, processed_at, status	Idempotency.
referrals	id, referrer_id, referred_id, code, status, reward_json	Growth loop.
coupons	id, code, type, value, max_redemptions, expires_at	Promo terkontrol.

AI, Notification & Audit

Table	Kolom kunci	Tujuan
ai_insights	id, user_id, enrollment_id, period, insight_type, content_json, model, prompt_version	Output terstruktur.
ai_usage	id, user_id, workspace_id, provider, model, input_tokens, output_tokens, cost_estimate	Budget control.
notifications	id, user_id, channel, type, payload_json, scheduled_at, sent_at, status	In-app/email/WA adapter.
audit_logs	id, workspace_id, actor_id, action, entity_type, entity_id, metadata_json, ip_hash	Forensik perubahan sensitif.
feature_flags	key, scope_type, scope_id, value_json, expires_at	Progressive rollout.

Index wajib

- workspace_members(user_id, workspace_id, status)
- activity_logs(enrollment_id, activity_date DESC)
- program_enrollments(user_id, status)
- subscriptions(workspace_id, status, current_period_end)
- notifications(status, scheduled_at)
- ai_usage(workspace_id, created_at)
- audit_logs(workspace_id, created_at DESC)

SECURITY

06. Auth, RLS & Security

Role model

Scope	Role	Hak utama
System	super_admin	Support internal; harus diaudit.
Workspace	owner	Billing, member, settings.
Workspace	admin	Program + member, tanpa ownership

Scope	Role	Hak utama
		transfer.
Workspace	coach	Client assignment, progress, intervention.
Workspace	member	Program dan data milik sendiri.
Read-only	viewer	Analytics tanpa edit.

RLS policy pattern

```
-- Pseudocode policy pattern
USING (
  EXISTS (
    SELECT 1 FROM workspace_members wm
    WHERE wm.workspace_id = target.workspace_id
      AND wm.user_id = auth.uid()
      AND wm.status = 'active'
  )
)
```

Security checklist

- Validasi Zod di boundary server untuk semua input.
- Service-role key hanya di server; jangan pernah NEXT_PUBLIC.
- Rate limit login-adjacent, invite, AI, export, dan webhook endpoints.
- Webhook: verify signature, store event_id, idempotent transaction.
- Ownership dan entitlement diperiksa server-side setiap mutation.
- Audit log untuk role change, billing, export, delete, dan admin impersonation.
- Export data dan delete account mengikuti retention policy yang jelas.
- AI context meminimalkan PII dan tidak mengirim catatan sensitif secara default.

Anti-pattern yang dilarang

Jangan mengandalkan hidden menu di frontend sebagai authorization. Jangan menyimpan role sebagai satu string di profile untuk semua tenant. Jangan menerima price/plan entitlement dari client.

HIGH-ARPU FEATURE

07. Coach Mode

Tujuan

Coach Mode mengubah Arva Tracker dari tool personal menjadi client success system. Coach tidak perlu mengecek WhatsApp dan spreadsheet satu per satu; dashboard memprioritaskan client yang membutuhkan perhatian.

Coach dashboard

Widget	Isi	Action
Needs Attention	Client dengan inactivity, completion turun, streak putus	Open client / send nudge
Active Clients	Jumlah client aktif vs limit plan	Manage clients
Completion	Average 7/30-day completion	View cohort
Interventions	Open follow-up dan overdue	Resolve / log note
Programs	Enrollment dan completion per program	Edit / duplicate

Risk score sederhana (MVP)

```

risk_score =
  inactivity_3d * 35
+ completion_below_50 * 30
+ streak_broken * 20
+ missed_priority_tasks * 15

0-29   = GREEN
30-59  = YELLOW
60-100 = RED

```

Gunakan deterministic rules dulu. AI hanya menjelaskan pola dan memberi saran wording; jangan membangun black-box risk scoring pada MVP.

Flow coach-client

1. Coach membuat workspace Coach.
2. Coach membuat/duplicate program.
3. Coach mengundang client melalui link/token.
4. Client join dan memberi consent progress sharing.
5. Coach melihat progress sesuai permission.
6. Sistem menandai risk state dan membuat suggested intervention.
7. Coach mengirim nudge dan mencatat intervention.

Fitur Coach Pro MVP

- Client list + search/filter
- Client detail + 7/30 day trend
- Needs Attention queue
- Program assignment
- Invite link
- Coach note private
- Nudge template
- CSV export ringan
- Plan limit + upgrade wall

Permission coach-client

Data	Coach dapat melihat?	Rule
Completion & streak	Ya	Hanya linked client aktif
Activity detail	Ya	Sesuai program yang ditugaskan
Reflection pribadi	Tidak default	Explicit consent / shared flag
Private coach note	Coach saja	Tidak pernah dikirim ke client
Billing client	Tidak	Hanya workspace owner/admin terkait

Intervention types

- Nudge: pengingat singkat untuk kembali ke ritme.
- Check-in: coach meminta update/context dari client.
- Review: evaluasi mingguan berdasarkan progress.
- Escalation: tandai follow-up manual tanpa otomatisasi agresif.

B2B2C

08. Community Mode

Konsep

Community Mode mengelola challenge atau cohort dengan banyak member. Fokusnya bukan membuka detail personal berlebihan, tetapi memberi admin aggregate progress, participation, dan mekanisme engagement.

Community dashboard

Metric	Definisi
Participants	Member terdaftar di challenge
Active Today	Member dengan ≥ 1 activity hari ini
Completion Rate	Completed steps / scheduled steps
7-day Active	Member aktif minimal 4 dari 7 hari
At Risk	Tidak aktif ≥ 3 hari atau completion $< 40\%$
Top Consistency	Leaderboard berbasis consistency score, bukan jumlah aktivitas mentah

Privacy rule

Aggregate by default

Admin komunitas melihat aggregate dan leaderboard sesuai consent. Catatan/reflection pribadi tidak pernah tampil kecuali user memilih share atau relationship coach-client memang mengizinkan.

Fitur MVP Community

- Create challenge dari program
- Invite link / code
- Member roster
- Progress aggregate
- Leaderboard opt-in
- Announcement / nudge
- CSV export
- Role owner/admin/coach/member
- Community branding ringan

AI LAYER

09. AI Coach



AI Coach menggunakan structured metrics, privacy filter, dan model routing untuk menjaga biaya dan kualitas.

Use case prioritas

Use case	Frekuensi	Model class	Output
Daily micro nudge	On-demand / rule-trigger	Fast/cheap	1-2 kalimat action-oriented
Weekly insight	Mingguan	Reasoning	Summary + pattern + next step
Coach client summary	On-demand	Fast/reasoning	Progress, risk factors, suggested intervention
Program copy assist	On-demand	Creative	Title, description, instruction
Classification	Background	Fast/cheap	Risk/intent/tag JSON

Model routing policy

- Fast/cheap model untuk classification, rewrite, dan nudge.
- Reasoning model hanya untuk weekly analysis atau complex coach summary.
- Creative model hanya saat user membuat program/copy.
- Provider abstraction melalui interface generateStructured() dan generateText().
- Set timeout, retry with jitter, fallback provider, dan hard token budget per plan.

Prompt contract

```

INPUT
- structured progress metrics
- user goal / program title
- timezone + period
- optional reflection summary (opt-in)
  
```

```

OUTPUT JSON
{
  "summary": "...",
  "wins": ["..."],
  "risks": ["..."],
  "next_actions": ["..."],
  "tone": "supportive"
}

```

AI safety & cost

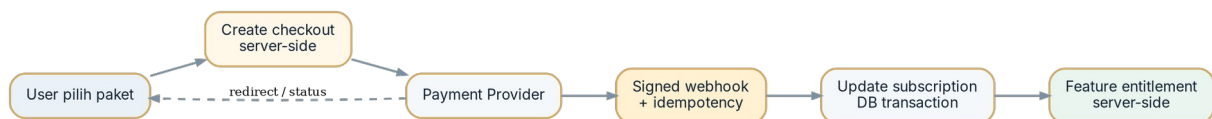
- Jangan kirim raw PII jika metrics cukup.
- Strip email, phone, token, dan secret dari context.
- Validate output schema sebelum disimpan.
- Store prompt_version dan model untuk audit/reproducibility.
- Per plan: daily token budget + monthly soft limit + fallback message jika budget habis.
- AI tidak boleh mengklaim diagnosis medis/psikologis dari pola kebiasaan.

Cost guardrail & acceptance

Guardrail	MVP rule
Weekly insight	Maks. 1 generated insight/user/week kecuali regenerate berbayar
Coach summary	On-demand dengan rate limit per workspace
Fallback	Jika provider gagal, tampilkan deterministic summary dasar
Cache	Gunakan insight tersimpan untuk periode yang sama
Quality	Output wajib lolos schema; invalid output tidak ditagihkan ke UX

BILLING

10. Payment Flow



Flow billing harus idempotent dan entitlement tidak boleh bergantung pada redirect browser.

Strategi provider

Gunakan interface PaymentProvider agar domain billing tidak terkunci. Implementasi pertama dapat menggunakan Midtrans atau Xendit sesuai merchant readiness. Semua price, plan, coupon, dan entitlement diambil dari server database.

Checkout sequence

1. User klik Upgrade.
2. Server validasi workspace ownership dan target plan.

3. Server menghitung price + coupon dari database.
4. Server membuat checkout/provider session dan menyimpan pending transaction.
5. User menyelesaikan pembayaran di provider.
6. Provider mengirim signed webhook.
7. Server cek idempotency event_id dan signature.
8. Dalam DB transaction: update transaction + subscription + entitlement cache jika ada.
9. Frontend membaca subscription dari server; redirect success hanya kosmetik.

Subscription states

```

trialing -> active -> past_due -> grace -> canceled
          \-> active (payment recovered)

pending_payment -> active
expired -> canceled

```

Billing failure UX

- Past due: banner lembut + tombol bayar ulang.
- Grace: fitur read tetap hidup; mutation premium tertentu dibatasi.
- Canceled: data tidak dihapus; downgrade entitlement pada period end.
- Provider timeout: tampilkan status pending, jangan double-create invoice.

GROWTH

11. Analytics, Retention & Growth

Event tracking minimal

Event	Trigger	Property pending
signup_completed	Onboarding selesai	source, campaign
program_created	Program baru	workspace_type
enrollment_started	Join program	program_id, source
activity_completed	Check item	step_id, date
week_success	>=70% weekly completion	rate, streak
paywall_viewed	Paywall tampil	feature, current_plan
checkout_started	Upgrade	plan, cycle
subscription_activated	Webhook success	plan, amount
coach_invite_accepted	Client join	workspace_id
ai_insight_viewed	Insight dibuka	type, model

Retention loops

- Daily: today list + progress ring + streak.

- Weekly: report dengan wins/risks/next action.
- Social: shareable achievement card.
- Accountability: partner/coach nudge.
- Coach: Needs Attention queue.
- Community: challenge milestone + leaderboard opt-in.

Referral design

Simple first

1 referral = 7 hari Pro, 3 referral = 30 hari Pro. Reward diposting sebagai ledger/credit record agar audit dan anti-abuse mudah.

EXPERIENCE

12. UI/UX Blueprint

Information architecture

```

/app
  /today          -> fokus hari ini
  /programs       -> program & enrollment
  /progress       -> trend, streak, report
  /insights       -> AI weekly insight
  /accountability -> partner / sharing
  /coach          -> coach dashboard (entitled)
  /community      -> community dashboard (entitled)
  /billing        -> plan, invoices, usage
  /settings       -> profile, workspace, privacy

```

Today screen

- Hero kecil: tanggal + progress hari ini.
- Primary action: activity list, maksimal 5-8 visible tanpa scroll berlebihan.
- Quick reflection hanya setelah progress minimum atau pada jam yang sesuai.
- Streak ditampilkan sebagai feedback, bukan hukuman.
- Empty state harus menjelaskan langkah berikutnya dan satu CTA.

Coach screen

- Default landing = Needs Attention, bukan chart.
- Risk chip green/yellow/red selalu punya alasan yang dapat dilihat.
- Bulk nudge dibatasi untuk mencegah spam.
- Client detail membedakan private coach note vs shared note.

Loading, error, offline

- Skeleton untuk dashboard blocks, bukan spinner full-screen.
- Optimistic update hanya untuk check/uncheck yang aman; rollback jika gagal.
- Error message manusiawi + retry; log technical context ke Sentry.
- PWA/offline optional Phase 2; jangan menahan MVP karena offline sync.

ENGINEERING

13. API & Background Jobs

Route handlers / server actions

Route/Action	Method	Guard
/api/billing/checkout	POST	auth + owner + rate limit
/api/billing/webhook/[provider]	POST	signature + idempotency
/api/invites/accept	POST	token + auth
/api/ai/insight	POST	auth + entitlement + budget
/api/export/progress	POST	auth + entitlement + audit
server action: completeActivity	mutation	ownership + validation
server action: sendCoachNudge	mutation	coach role + consent + rate limit

Background jobs

Job	Schedule	Output
Generate weekly snapshot	Daily/hourly by timezone	progress_snapshots
Generate weekly AI insight	Weekly after snapshot	ai_insights
Send scheduled reminders	Every 5-15 min	notifications
Subscription reconciliation	Daily	repair drift
Stale invite cleanup	Daily	expire invites
Audit retention / cleanup	Monthly	retention policy

Gunakan cron/queue yang tersedia di deployment saat ini. Bungkus job dalam idempotent handler sehingga bisa dipindahkan ke queue dedicated tanpa mengubah domain logic.

RELIABILITY

14. Observability & Operations

Minimum production baseline

- Sentry error + performance traces pada server dan client.
- Structured log dengan request_id, workspace_id (non-sensitive), action, latency.
- Health endpoint untuk DB + provider config (tanpa membocorkan secret).
- DB backup dan restore drill minimal bulanan setelah monetisasi aktif.
- Feature flags untuk AI Coach, Coach Mode, Community Mode.
- Admin support tools read-only by default dan setiap impersonation diaudit.

SLO awal

Area	Target
Core dashboard p95	< 2.0s pada koneksi normal
Mutation p95	< 700ms selain provider/AI
Error rate core	< 1%
Webhook processing	99% < 60 detik
AI weekly job success	> 98% setelah retry
Availability	99.5% MVP → 99.9% ketika B2B besar

SHIPPING PLAN

15. Roadmap MVP 90 Hari

Milestone 12 minggu

Minggu	Fokus	Deliverable release
1	Audit + foundation	Repo audit, ADR, schema baseline, design tokens, env validation
2	Workspace + RLS	Personal workspace, member roles, RLS tests
3	Program V2	Program/steps/enrollment migration + UI
4	Tracking core	Today, activity logs, progress snapshot, analytics events
5	Billing foundation	Plans, entitlements, checkout adapter, webhook idempotency
6	Personal Pro	Paywall, advanced progress, export, annual plan
7	Coach Mode 1	Coach workspace, client invite, client list/detail
8	Coach Mode 2	Risk rules, Needs Attention, intervention/nudge
9	AI Coach	AI gateway, weekly insight, usage budget, fallback
10	Community Mode	Challenge, members, aggregate progress, leaderboard opt-in
11	Hardening	Security review, load test, observability, backup/restore
12	Launch	Pricing page, onboarding, docs, beta rollout, KPI dashboard

Release gates

Gate 1 - Week 4

Core tracking harus stabil dan RLS tests lulus sebelum billing masuk.

Gate 2 - Week 6

Billing webhook, idempotency, downgrade/upgrade, dan entitlement tests wajib lulus sebelum menerima uang.

Gate 3 - Week 9

AI hanya dilepas jika usage budget, schema validation, privacy filter, dan fallback bekerja.

Gate 4 - Week 12

Security regression, restore test, analytics, dan support playbook selesai sebelum public launch.

Scope yang sengaja ditunda

- Marketplace creator payout
- Full white-label custom domain
- Complex gamification economy
- Native iOS/Android
- Realtime chat
- Advanced offline-first sync
- ML-based predictive churn

QUALITY

16. Definition of Done & KPI

Definition of Done fitur

- Acceptance criteria jelas dan teruji.
- Server-side validation + authorization.
- Loading/empty/error/success states.
- Analytics event untuk funnel penting.
- Unit/integration test pada domain kritis.
- RLS test jika menyentuh data tenant.
- Audit log jika mutation sensitif.
- No secret di client bundle.
- Mobile responsive dan keyboard accessible.
- Sentry breadcrumb/context yang cukup tanpa PII.

KPI launch dashboard

Funnel	KPI
Acquisition	Visitor → signup
Activation	Signup → first program → first check-in
Engagement	DAU/WAU, activities/user/week
Value	Weekly Successful Progress
Retention	D7, D30 cohort retention
Monetization	Free→Pro, trial→paid, MRR, ARPU, churn
Coach	Invites accepted, active clients/coach, intervention resolution
Community	Join rate, active rate, challenge completion
AI	Insight view rate, thumbs up/down, cost per active paid user

READY TO COPY-PASTE

17. Master Prompt Codex

Prompt berikut sengaja dibuat sebagai instruksi implementasi bertahap. Codex harus mengaudit repository lebih dulu dan menjaga kompatibilitas dengan aplikasi yang sudah live.

MASTER PROMPT - ARVA TRACKER V2

Anda adalah Principal Full Stack Engineer yang bertugas meng-upgrade repository Arva Tracker yang sudah ada menjadi SaaS V2 production-ready.

TUJUAN PRODUK

Arva Tracker adalah "Progress & Accountability Platform" dengan tagline "Langkah kecil, perubahan besar." Target persona: Individual, Coach/Mentor, Community, dan Business. Jangan mengubah produk menjadi sekadar habit tracker generik.

PRINSIP EKSEKUSI

1. Audit repository terlebih dahulu. Jangan langsung rewrite.
2. Deteksi framework, package manager, database, auth, deployment, dan convention yang sudah dipakai.
3. Pertahankan landing page/brand visual yang sudah baik. Refactor hanya jika ada alasan teknis/UX yang jelas.
4. Pilih solusi paling sederhana yang production-safe. Jangan over-engineer.
5. Semua auth, role, billing entitlement, ownership, dan harga diputuskan server-side.
6. Semua input divalidasi dengan schema validation (prefer Zod jika stack kompatibel).
7. Tidak boleh ada secret/service-role key pada client bundle.
8. Setiap perubahan schema harus melalui migration yang reversible/terdokumentasi.
9. Semua halaman baru wajib mobile responsive, loading state, empty state, error state, dan accessibility dasar.
10. Jangan menghapus data existing. Buat migration/backfill yang aman.

STACK TARGET

- Next.js App Router + TypeScript (gunakan versi existing atau latest stable yang kompatibel; jangan upgrade major tanpa alasan).
- Tailwind CSS untuk styling dan design tokens.
- Supabase Auth + PostgreSQL + RLS jika repository memang menggunakan Supabase; jika DB berbeda, pertahankan adapter tetapi ikuti domain model di bawah.
- PaymentProvider abstraction dengan implementasi Midtrans atau Xendit berdasarkan ENV/config yang tersedia.
- AIProvider abstraction agar bisa menggunakan OpenAI-compatible/Anthropic/provider lain tanpa

mengikat domain.

- Sentry atau observability existing.
- Vitest/Jest untuk unit/integration dan Playwright untuk critical E2E jika toolchain kompatibel.

DESIGN SYSTEM

Pertahankan identitas Arva Tracker: dark navy, white, elegant gold accent. Dashboard app boleh menggunakan surface terang/dark sesuai desain existing, tetapi harus konsisten. Buat design token, jangan hard-code warna berulang.

FASE 0 - AUDIT

Sebelum coding:

- Print ringkasan arsitektur repository.
- Identifikasi package manager dan commands.
- Identifikasi route, auth, schema, migrations, env, deployment.
- Cari technical debt yang menghalangi V2.

- Buat file docs/ARVA_TRACKER_V2_PLAN.md berisi audit, keputusan, risiko, dan urutan migration.
- Jangan melakukan destructive change.

DOMAIN MODEL WAJIB

Gunakan auth provider untuk identity, lalu buat domain minimum:

- profiles
- workspaces (type: personal|coach|community|business)
- workspace_members (role: owner|admin|coach|member|viewer)
- workspace_invites
- programs
- program_steps
- program_enrollments
- activity_logs
- daily_reflections
- progress_snapshots
- coach_client_links
- accountability_links
- coach_interventions
- community_challenges
- challenge_members
- plans
- subscriptions
- billing_transactions
- webhook_events
- notifications
- ai_insights
- ai_usage
- audit_logs
- feature_flags
- referrals (boleh Phase 2)

DATABASE RULES

- Primary key UUID.
- timestamptz UTC.
- FK + ON DELETE behavior eksplisit.
- Unique constraint untuk membership/enrollment yang relevan.
- Index untuk workspace membership, activity logs by enrollment/date, active subscription, scheduled notifications, audit logs.

- JSONB hanya untuk flexible config/metrics; data yang sering di-query harus kolom relational.
- Personal workspace otomatis dibuat untuk setiap user.
- Semua collaborative entity memiliki workspace_id secara langsung atau bisa diturunkan secara aman.

RLS / AUTHORIZATION

Jika Supabase:

- Enable RLS pada semua tabel user/tenant data.
- Policy membaca workspace membership dari auth.uid().

- User hanya melihat data pribadi kecuali relation/permission eksplisit.
 - Coach hanya melihat client yang linked/assigned.
 - Community admin melihat aggregate/member progress sesuai permission; reflection private tidak boleh bocor.
 - Service role hanya server-side.
- Tambahkan test yang membuktikan cross-tenant read/write ditolak.

PRICING & ENTITLEMENT

Seed plan:

1. FREE - Rp0
2. PERSONAL_PRO - Rp39.000/bulan, Rp349.000/tahun
3. COACH_PRO - Rp199.000/bulan, Rp1.790.000/tahun
4. COMMUNITY - Rp599.000/bulan, Rp5.390.000/tahun
5. BUSINESS - custom/manual

Entitlement disimpan sebagai config server-side dari plan, contoh:

- max_active_programs
- history_days
- advanced_analytics
- ai_weekly_insights
- max_accountability_partners
- max_clients
- max_community_members
- export_enabled
- custom_branding

Jangan hard-code limit di UI. Buat helper server `getEntitlements(workspaceId)`.

BILLING

Buat interface `PaymentProvider`:

- `createCheckout()`

- `verifyWebhook()`
- `parseEvent()`
- `getPaymentStatus()` bila perlu

Implement provider berdasarkan env. Jika credential belum ada, buat mock/dev provider yang aman untuk local development tanpa meniru webhook production secara insecure.

Flow wajib:

1. POST checkout server-side.
2. Price/plan/coupon dihitung server-side.
3. Simpan pending billing transaction.
4. Verify webhook signature.
5. Idempotency via provider `event_id`.
6. Dalam DB transaction update transaction + subscription.
7. Entitlement dibaca dari subscription server-side.
8. Redirect browser tidak boleh mengaktifkan subscription.

Status subscription minimum:

trialing, active, past_due, grace, canceled, expired.

Support upgrade, downgrade effective period, cancellation, and failed payment UX.

PERSONAL PRO

Implement:

- unlimited programs (sesuai entitlement)
- unlimited history
- advanced progress 7/30/90 days
- weekly report
- export CSV/PDF sederhana
- paywall component reusable
- plan/billing page

COACH MODE

Routes/UI minimum:

```
/app/coach
/app/coach/clients
/app/coach/clients/[id]
/app/coach/programs
```

Fitur:

- Coach workspace.
- Invite client dengan signed/random token; simpan token hash, expiry.
- Client accept dengan consent.
- Client list + search/filter.
- Client detail: 7/30-day progress, streak, recent activity, program enrollment.
- Needs Attention queue.
- Deterministic risk score MVP:
 - inactivity $\geq 3d$ = +35
 - completion 7d $< 50\%$ = +30
 - streak broken = +20
 - priority tasks missed = +15
 - 0-29 green, 30-59 yellow, 60-100 red.
- Tampilkan reason breakdown untuk risk score.
- Coach intervention log.
- Nudge template; rate limit dan audit log.
- Private coach note tidak terlihat client kecuali explicitly shared.

COMMUNITY MODE

Routes minimum:

```

/app/community
/app/community/challenges
/app/community/challenges/[id]
/app/community/members

```

Fitur:

- Create challenge from program.
- Invite link/code.
- Member roster.
- Aggregate progress.
- Metrics: participants, active today, completion rate, 7-day active, at-risk.
- Leaderboard opt-in; jangan expose data private.
- Announcement/nudge.
- Role owner/admin/coach/member.
- Light community branding gated by entitlement.

AI COACH

Buat ai gateway server-only:

- AIProvider interface.
- model router: FAST dan REASONING.
- configurable provider/model via ENV.
- timeout + retry + fallback.
- output schema validation.
- prompt_version dan model disimpan.
- ai_usage simpan input/output token dan cost estimate jika tersedia.
- per-plan usage limit/soft budget.

MVP AI use case:

1. Weekly Personal Insight.
2. Coach Client Summary on-demand.
3. Optional Daily Micro Nudge.

Privacy:

- Default input menggunakan structured metrics, bukan raw notes.
- Redact email/phone/token/secret dari context.
- Reflection hanya dikirim bila explicit opt-in/config.
- Jangan membuat diagnosis medis/psikologis.

Weekly insight output JSON:

```

{
  "summary": string,
  "wins": string[],

```

```

    "risks": string[],
    "next_actions": string[],
    "tone": "supportive"
  }

```

Render dengan UI card yang clean dan action-oriented.

BACKGROUND JOBS

Buat idempotent job handlers:

- progress snapshot generation
- weekly AI insight
- scheduled reminders
- subscription reconciliation
- invite expiration cleanup

Jangan mengikat domain ke satu scheduler. Buat endpoint/job function yang bisa dipanggil Vercel Cron, server cron, atau queue.

ANALYTICS EVENTS

Buat analytics adapter. Track minimum:

- signup_completed
- program_created
- enrollment_started
- activity_completed
- week_success
- paywall_viewed
- checkout_started
- subscription_activated
- coach_invite_accepted
- ai_insight_viewed

Jangan memasukkan note/private text sebagai event property.

OBSERVABILITY

- Structured error handling.
- request_id/correlation id.
- Sentry existing jika tersedia.
- audit_logs untuk role change, billing, export, delete, coach intervention sensitif, dan admin support action.
- Jangan log secret/token/full payment payload.

PERFORMANCE

- Server Components default; use client hanya untuk interaksi.
- Avoid N+1 queries pada dashboard coach/community.
- Index berdasarkan query nyata.
- Lazy load chart/AI panel berat.
- Paginate member/client list.
- Cache aggregate yang aman melalui progress_snapshots bila query mahal.
- Image optimization dan bundle check.

UX REQUIREMENTS

- Mobile-first.
- Skeleton untuk dashboard, bukan spinner global.
- Error state dengan retry.
- Empty state punya satu CTA yang jelas.

- Optimistic update hanya untuk activity check/uncheck dan harus rollback ketika gagal.
- Paywall tidak menghapus data; hanya membatasi premium actions.
- Accessibility: label, focus state, keyboard navigation, contrast.

FOLDER ORGANIZATION (ADAPT TO REPO)

Prefer struktur domain-based, misalnya:

```

src/
  app/
  components/
  features/
  auth/

```



```

    programs/
    tracking/
    billing/
    coach/
    community/
    ai/
  lib/
    db/
    auth/
    payments/
    ai/
    analytics/
    security/
  server/
    services/
    jobs/
    types/
    validation/
  supabase/
    migrations/
    seed.sql
  docs/

```

Jangan memindahkan seluruh repo hanya demi mengikuti struktur ini. Refactor incremental.

TEST WAJIB

1. Auth/authorization:
 - cross tenant read denied
 - cross tenant write denied
 - coach cannot see unrelated client
2. Billing:
 - webhook signature invalid -> reject
 - duplicate event -> no duplicate activation
 - success -> subscription active
 - canceled/expired -> entitlement downgrade
3. Tracking:
 - activity completion ownership
 - weekly progress calculation
4. Coach:
 - risk score deterministic
 - invitation expiry/acceptance
5. AI:
 - output schema validation
 - budget exceeded fallback
 - provider timeout fallback
6. Critical E2E:
 - signup -> create program -> check activity -> upgrade mock -> access Pro
 - coach -> invite client -> client accepts -> coach sees progress

MIGRATION STRATEGY

- Inspect current schema and map old table -> new domain.
- Add new tables/columns first.
- Backfill in separate migration/script.
- Dual-read only jika benar-benar perlu.
- Jangan drop old columns sampai compatibility terverifikasi.
- Buat docs migration notes.

ROADMAP EKSEKUSI

Implement bertahap:

- A. Foundation: workspace/RLS/domain migrations.
- B. Program/tracking V2.
- C. Billing + Personal Pro.
- D. Coach Mode.

E. AI Coach.
F. Community Mode.
G. Hardening + launch.

SETIAP FASE HARUS:

- jelaskan file yang akan diubah
- implement code lengkap
- jalankan lint/typecheck/test/build
- perbaiki error sampai bersih
- update docs/ARVA_TRACKER_V2_PLAN.md
- commit-ready output (jangan commit kecuali diminta)

DEFINITION OF DONE

Fitur belum selesai jika:

- auth hanya di frontend
- tidak ada validation
- tidak ada loading/error/empty state
- RLS/test tenant tidak ada untuk data sensitif
- billing mengandalkan redirect
- webhook tidak idempotent
- AI output tidak divalidasi
- TypeScript/lint/build gagal

LANGKAH PERTAMA SEKARANG

1. Audit repository secara menyeluruh.
2. Tampilkan ringkasan temuan dan gap terhadap blueprint ini.
3. Buat docs/ARVA_TRACKER_V2_PLAN.md.
4. Usulkan migration plan paling aman.
5. Setelah itu mulai FASE A tanpa meminta saya menyalin ulang konteks, kecuali ada keputusan bisnis yang benar-benar blocking.

Expected output pertama dari Codex

- Repository audit: stack, route, schema, auth, deployment, dan technical debt.
- Gap analysis terhadap blueprint V2 tanpa melakukan destructive rewrite.
- docs/ARVA_TRACKER_V2_PLAN.md berisi ADR, migration strategy, risiko, dan urutan fase.
- Daftar file/migration yang akan dibuat pada Fase A serta command test/build yang akan dijalankan.
- Implementasi Fase A baru dimulai setelah audit selesai dan repository tetap buildable.

Rule untuk sesi coding

Jika Codex menemukan stack existing berbeda dari asumsi blueprint, pertahankan komponen yang sehat dan adaptasikan domain model. Jangan melakukan major upgrade hanya untuk mengejar versi terbaru.

REFERENCE

18. Appendix Implementasi

Environment variables

```
# Public
NEXT_PUBLIC_APP_URL=
NEXT_PUBLIC_SUPABASE_URL=
NEXT_PUBLIC_SUPABASE_ANON_KEY=

# Server only
SUPABASE_SERVICE_ROLE_KEY=
```

```

PAYMENT_PROVIDER=midtrans|xendit
PAYMENT_SERVER_KEY=
PAYMENT_WEBHOOK_SECRET=
AI_PROVIDER=
AI_FAST_MODEL=
AI_REASONING_MODEL=
AI_API_KEY=
SENTRY_DSN=
CRON_SECRET=

```

```

# Optional
EMAIL_PROVIDER=
EMAIL_API_KEY=
RATE_LIMIT_REDIS_URL=
RATE_LIMIT_REDIS_TOKEN=

```

Feature flags launch

Flag	Default	Rollout
billing_v2	off → internal → 10% → 100%	Pastikan webhook stabil
coach_mode	internal → beta coaches → 100%	Pantau invite acceptance
ai_weekly_insight	internal → Pro 25% → 100%	Pantau cost + feedback
community_mode	invite-only	Naikkan setelah aggregate query stabil
referral_rewards	off	Aktif setelah anti-abuse siap

Folder tree rekomendasi

```

src/
├─ app/
│   └─ (marketing)/
│       └─ app/
│           ├── today/
│           ├── programs/
│           ├── progress/
│           ├── coach/
│           ├── community/
│           └─ billing/
├─ features/
│   ├── programs/
│   ├── tracking/
│   ├── billing/
│   ├── coach/
│   ├── community/
│   └─ ai/
├─ lib/
│   ├── auth/
│   ├── db/
│   ├── payments/
│   ├── ai/
│   ├── analytics/
│   └─ security/
├─ server/
│   ├── services/
│   └─ jobs/
└─ validation/

supabase/

```

```
├─ migrations/
├─ seed.sql

docs/
├─ ARVA_TRACKER_V2_PLAN.md
```

Launch checklist

- Domain/SSL production
- Transactional email verified
- Payment live keys + webhook verified
- Privacy Policy + Terms + refund/cancellation policy
- RLS regression test
- Backup + restore test
- Sentry alert route
- Analytics dashboard
- Support email/WhatsApp channel
- Beta cohort dan rollback plan

Keputusan yang sebaiknya tidak ditunda sebelum coding

Decision	Default rekomendasi
Primary paid audience	Coach Pro setelah Personal Pro
Billing cycle	Monthly + annual
Provider	Adapter; pilih merchant ready provider
Community leaderboard	Opt-in
AI reflection usage	Off by default, opt-in
Data deletion grace	30 hari sebelum hard delete backup
Business plan	Manual sales + custom entitlement

KICKOFF STRATEGY

Mulai V2 dari bukti nilai, bukan jumlah fitur.

Final recommendation

Ship V2 secara bertahap. Jangan menunggu semua fitur selesai. Target pertama adalah membuktikan bahwa pengguna mau membayar untuk progress visibility dan coach mau membayar untuk client accountability. Setelah recurring revenue dan retention terbukti, barulah marketplace dan white-label menjadi pengungkit valuation.

3 tindakan pertama setelah dokumen ini

1. Submit Master Prompt Codex ke repository Arva Tracker dan lakukan audit tanpa destructive change.
2. Validasi schema + RLS foundation terlebih dahulu; billing baru masuk setelah isolation test lulus.
3. Rekrut 5-10 beta coach dan ukur invite acceptance, client weekly progress, serta willingness-to-pay.

Target outcome 90 hari

Outcome	Bukti sukses
Personal Pro	Checkout + entitlement aktif dan ada pengguna berbayar
Coach Pro	Coach mengundang client dan memakai Needs Attention
AI Coach	Weekly insight stabil dengan cost guardrail
Community	Challenge beta berjalan dengan aggregate progress
Production	RLS, webhook idempotency, observability, backup/restore lulus

North Star

Arva Tracker bernilai tinggi ketika progress menjadi data yang dipercaya, accountability menjadi workflow yang dipakai setiap minggu, dan revenue berulang tumbuh tanpa mengorbankan keamanan atau UX.